RIPHAH INTERNATIONAL UNIVERSITY



Artificial Intelligence

LAB #11

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Task 1:

Write Prolog facts for the following statements: (a) Ali likes apples. (b) Sara is a student. (c) The sky is blue. (d) Ahmed owns a book. Save the program as facts.pl.

Task 2:

Load facts.pl in SWI-Prolog and run queries to determine: (a) What does Ali like? (b) Who is a student? (c) What does Ahmed own? Record the query and its output.

Task 3:

Use variables to ask Prolog questions such as likes(Who, apples) and owns(Who, What). Explain what each variable represents and write down the answers returned by Prolog.

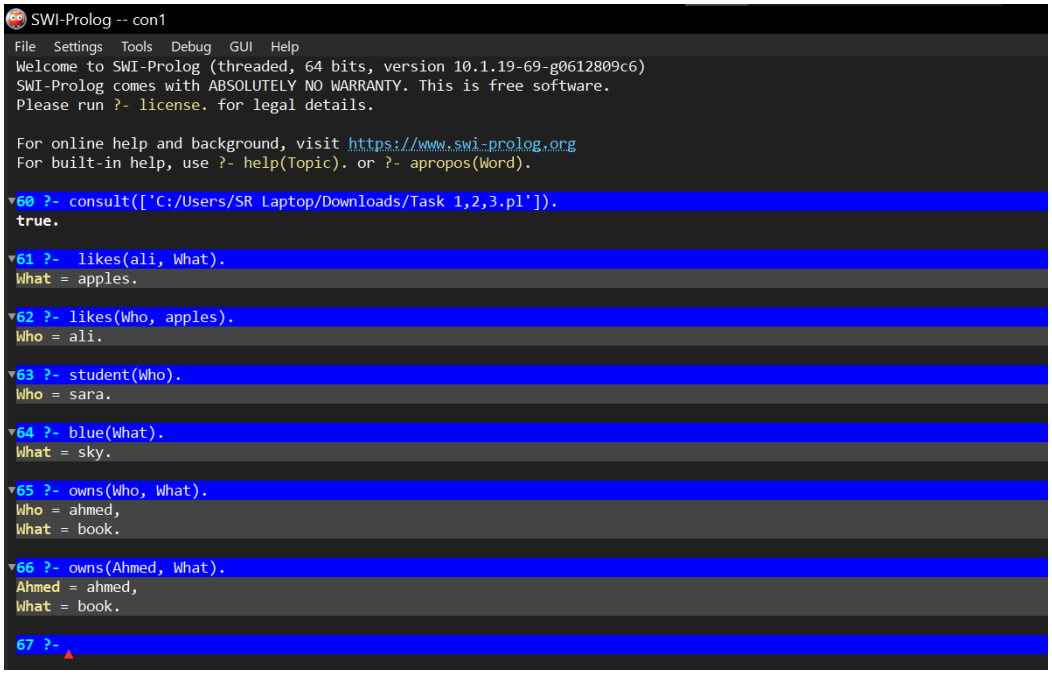
Prologs:

likes(ali, apples).

student(sara).

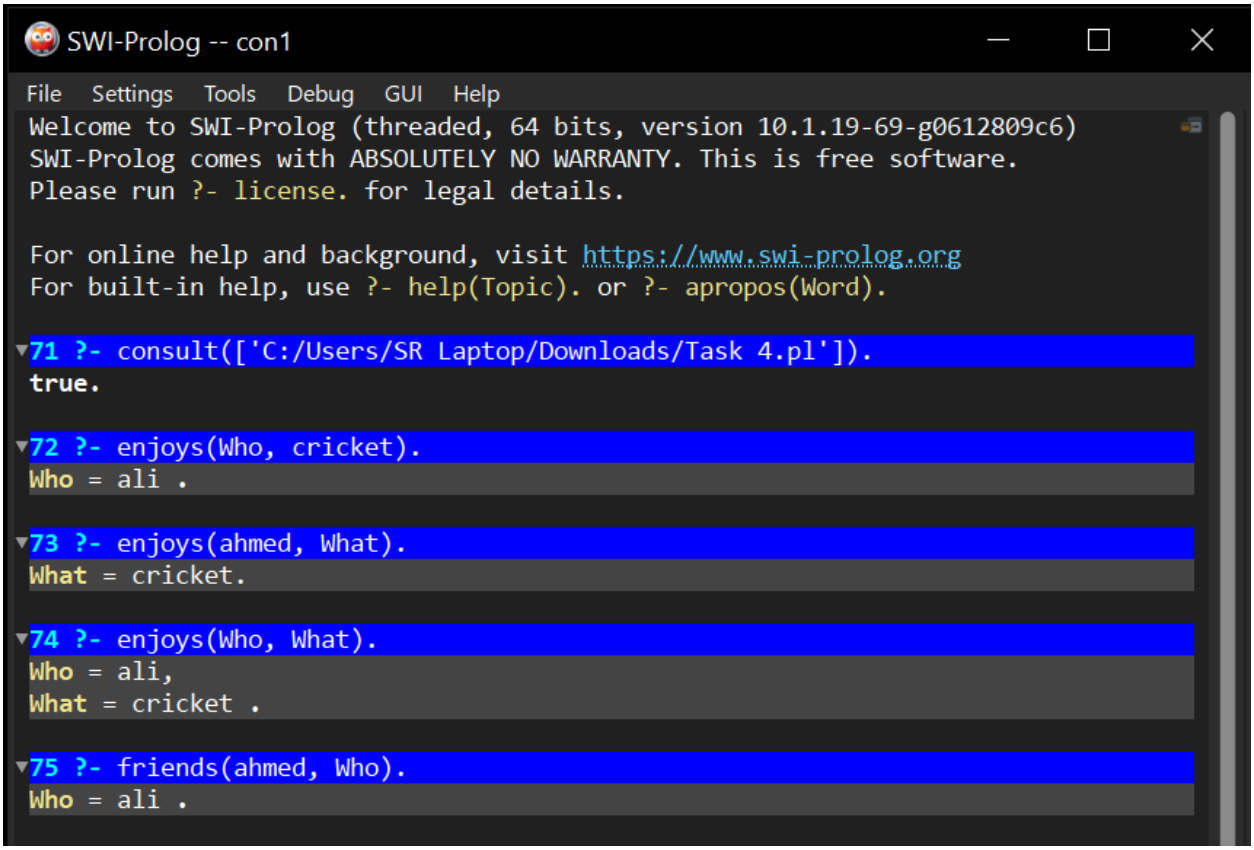
blue(sky).

owns(ahmed, book).



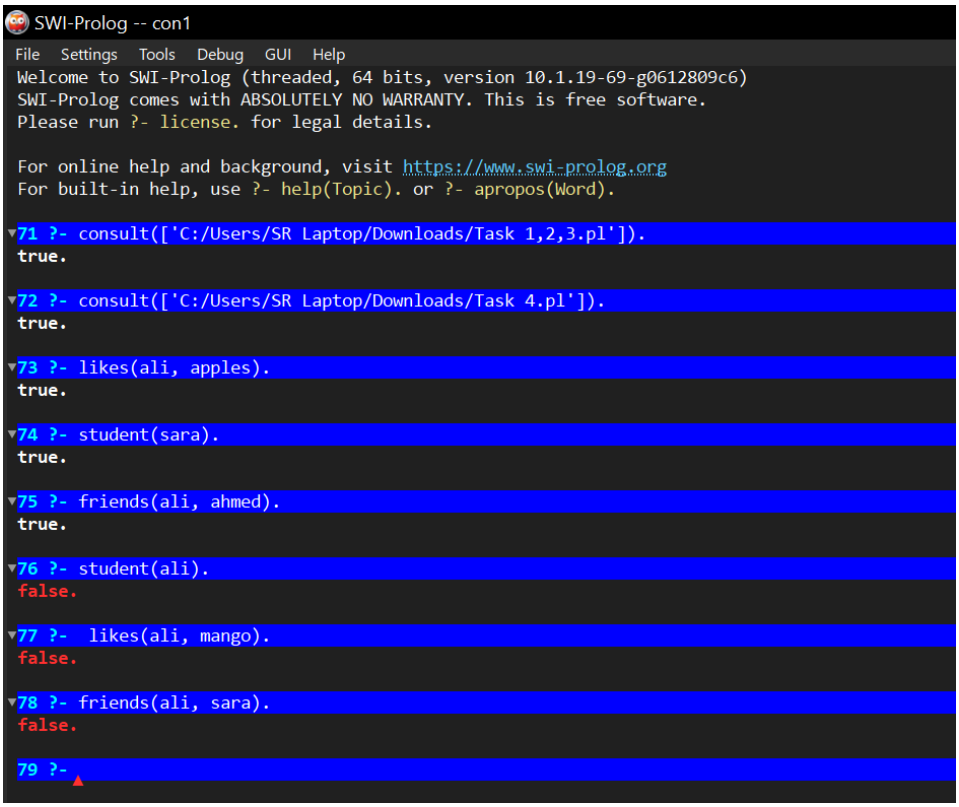
Task 4:

Create facts for two people who enjoy playing cricket. Write a rule that determines when two people are friends based on both enjoying cricket. Test the rule with suitable queries.



Task 5:

Run queries that should return true and queries that should return false. Write at least two examples of each and briefly explain why Prolog returns those results.



True examples:

- likes(ali, apples). is true because this exact fact is in the program.
- student(sara). is true for the same reason.
- friends(ali, ahmed). is true because the rule's conditions are all met. Both Ali and Ahmed enjoy cricket, and they are different people.

False examples:

- student(ali). is false because no fact says Ali is a student.
- likes(ali, mango). is false because the only fact about Ali's likes is likes(ali, apples).
- friends(ali, sara). is false because Sara doesn't enjoy cricket, so the rule's second condition fails.

Task 6:

Intentionally remove a period from one fact or change a predicate name. Run the program, identify the error or unexpected result, correct it, and describe what you learned.

